

MCQ and Coding Test (All Mandatory) – 1 Marks Each

1. What is a variable in Python?
 - A) A constant value
 - B) A named storage location to hold data
 - C) A reserved keyword
 - D) A data types
2. Which of the following is considered a constant in Python?
 - A) True
 - B) False
 - C) $\pi = 3.14$
 - D) None
3. What are the primitive data types in Python?
 - A) Integer, Float, String, Boolean
 - B) Number, Character, Boolean
 - C) Float, String, List, Tuple, Dictionary
 - D) Integer, Character, List, Tuple
4. What does the term "operands" refer to in Python?
 - A) Functions that operate on data
 - B) Values or variables on which operators perform operations
 - C) Data types
 - D) Reserved keywords
5. How can you swap the values of two variables a and b without using a temporary variable in Python?
 - A) $a = b, b = a$
 - B) $a = a + b, b = a - b, a = a - b$
 - C) $a, b = b, a$
 - D) $a = a * b, b = a / b, a = a / b$
6. What is the purpose of the "if" condition in Python?
 - A) To execute a block of code only if a certain condition is true
 - B) To execute a block of code regardless of any condition
 - C) To execute a block of code if a certain condition is false
 - D) To terminate the program
7. When is the "else" condition used in Python?
 - A) When there is only one possible outcome
 - B) When there are two possible outcomes
 - C) When there are multiple possible outcomes
 - D) When the "if" condition is false

8. What is the purpose of the "elif" condition in Python?
- A) To execute a block of code if the "if" condition is true
 - B) To execute a block of code if none of the previous conditions are true
 - C) To execute a block of code if the "if" condition is false and another condition is true
 - D) To terminate the program
9. How does the "while" loop differ from the "for" loop in Python?
- A) "while" loop is used for iteration over a sequence, "for" loop is used for indefinite repetition
 - B) "while" loop is used for indefinite repetition, "for" loop is used for iteration over a sequence
 - C) "while" loop is used for iteration over a sequence, "for" loop is used for iteration over a range of numbers
 - D) "while" loop is used for iteration over a range of numbers, "for" loop is used for indefinite repetition
10. What does the "break" statement do in a "for" loop in Python?
- A) Exits the loop entirely
 - B) Skips the current iteration and continues with the next iteration
 - C) Breaks the loop if a certain condition is met
 - D) None of the above
11. Which of the following statements is true about lists in Python?
- A) Lists are immutable
 - B) Lists can only contain elements of the same data type
 - C) Lists are ordered collections of items
 - D) Lists cannot be nested
12. What is the purpose of the enumerate function in Python?
- A) To return the sum of all elements in a list
 - B) To iterate over the index-value pairs of a list
 - C) To check if a certain element exists in a list
 - D) To reverse the order of elements in a list
13. What is an advantage of tuples over lists in Python?
- A) Tuples are mutable
 - B) Tuples can only store elements of the same data type
 - C) Tuples are ordered collections
 - D) Tuples are immutable
14. How do you delete a tuple in Python?
- A) Using the remove() method
 - B) Using the delete() function
 - C) Tuples cannot be deleted
 - D) Using the del keyword

15. What is the result of the union of two sets in Python?
- A) Intersection of the sets
 - B) Difference of the sets
 - C) Combination of the sets without duplicates
 - D) Merging the sets into one
16. What is a frozenset in Python?
- A) A set that cannot be modified after creation
 - B) A set with frozen elements
 - C) A set with only one element
 - D) A set that contains frozen data
17. How do you create a dictionary in Python?
- A) Using square brackets []
 - B) Using parentheses ()
 - C) Using curly braces { }
 - D) Using angle brackets < >
18. What does the items() method return in a dictionary?
- A) List of keys
 - B) List of values
 - C) List of key-value pairs
 - D) Number of key-value pairs
19. What is a variable in Python?
- A) A constant value
 - B) A container to store data
 - C) A reserved keyword
 - D) A data type identifier
20. Which of the following is a constant in Python?
- A) True
 - B) 5
 - C) "Hello"
 - D) None
21. Operators and Operands:
3. What are operands in Python?
- A) Functions that operate on data
 - B) Values or variables involved in operations
 - C) Data types
 - D) Reserved keywords
22. How can you swap the values of two variables a and b in Python?
- A) a = b, b = a
 - B) a = a + b, b = a - b, a = a - b
 - C) a, b = b, a
 - D) a = a * b, b = a / b, a = a / b

23. What is the purpose of the "if" condition in Python?
- A) To execute a block of code only if a certain condition is true
 - B) To execute a block of code regardless of any condition
 - C) To execute a block of code if a certain condition is false
 - D) To terminate the program
24. When is the "else" condition used in Python?
- A) When there is only one possible outcome
 - B) When there are two possible outcomes
 - C) When there are multiple possible outcomes
 - D) When the "if" condition is false
25. How does the "while" loop differ from the "for" loop in Python?
- A) "while" loop is used for indefinite repetition, "for" loop is used for iteration over a sequence
 - B) "while" loop is used for iteration over a sequence, "for" loop is used for indefinite repetition
 - C) "while" loop is used for iteration over a sequence, "for" loop is used for iteration over a range of numbers
 - D) "while" loop is used for iteration over a range of numbers, "for" loop is used for indefinite repetition
26. What does the "break" statement do in a "for" loop in Python?
- A) Exits the loop entirely
 - B) Skips the current iteration and continues with the next iteration
 - C) Breaks the loop if a certain condition is met
 - D) None of the above
27. Which of the following statements is true about lists in Python?
- A) Lists are immutable
 - B) Lists can only contain elements of the same data type
 - C) Lists are ordered collections of items
 - D) Lists cannot be nested
28. What is the purpose of the enumerate function in Python?
- A) To return the sum of all elements in a list
 - B) To iterate over the index-value pairs of a list
 - C) To check if a certain element exists in a list
 - D) To reverse the order of elements in a list

Coding Questions (Any 2) – 5 Marks Each

- Write a Python program that takes an integer as input and prints whether it is even or odd.
 - Output:
7 is odd.
- Write a Python program that takes a person's age as input and prints whether they are eligible to vote (considering the legal voting age as 18).
 - Output:
"You are eligible to vote".
- Write a Python program that takes a list of numbers as input and prints the sum of all the numbers.
 - Input: List : [5, 8, 12, 3, 9]
 - Output:
Sum of the numbers is 37
- Write a Python program that takes a list of strings as input and prints the longest string in the list.
 - List: [apple, banana, orange, strawberry]
 - Output: "Longest string is strawberry"
- Write a Python program that takes a tuple of numbers as input and prints the product of all the numbers.
 - Input:
Enter numbers separated by comma: 2, 3, 4, 5
 - Output:
Product of the numbers is 120
- Write a Python program that takes two tuples of equal length as input and prints a new tuple containing the element-wise sum of the two tuples.
 - Input:
Enter first tuple: (1, 2, 3)
Enter second tuple: (4, 5, 6)
 - Output:
Result tuple: (5, 7, 9)

- Write a Python program that takes a dictionary of student names and their respective scores as input and prints the name of the student with the highest score.
 - Input:
Enter student names and scores (name: score): Alice: 85, Bob: 90, Charlie: 78, David: 92
 - Output:
"Student with the highest score is David".

- Write a Python program that takes two dictionaries as input, where each dictionary represents the inventory of items in two different stores. Print the items that are common in both stores.
 - Input:
Enter inventory of store 1 (item: quantity): apple: 10, banana: 5, orange: 8
Enter inventory of store 2 (item: quantity): banana: 3, orange: 6, strawberry: 7
 - Output:
Common items in both stores: banana, orange

- Write a Python program that takes two sets of integers as input and prints the union of the two sets.
 - Input:
Enter integers for set 1 (separated by comma): 1, 2, 3, 4
Enter integers for set 2 (separated by comma): 3, 4, 5, 6
 - Output:
Union of the two sets: {1, 2, 3, 4, 5, 6}

- Write a Python program that takes a set of words as input and prints the unique vowels present in all the words.
 - Input:
Enter words separated by space: apple banana orange strawberry
 - Output:
Unique vowels in the words: {'e', 'a', 'o'}

----- **END** -----